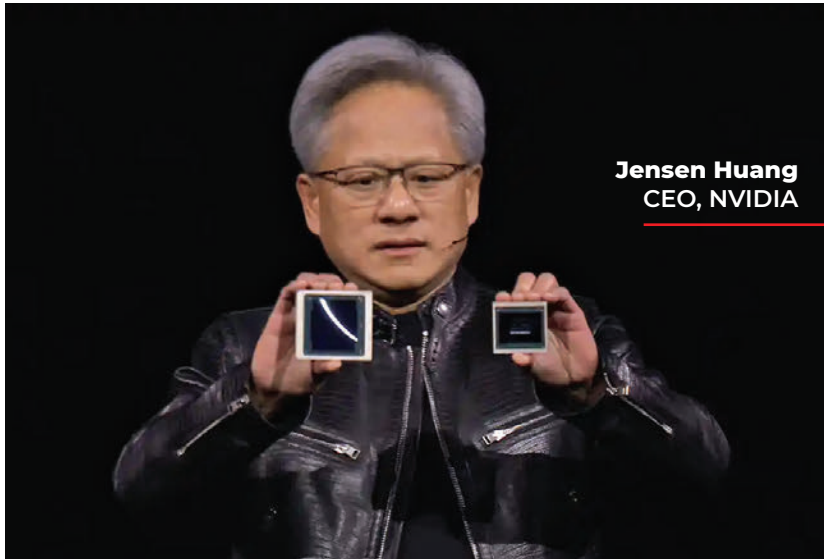




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GTC 2024

NVIDIA Blackwell, GB200, NVL72, SuperPODs,
and more revealed

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NVIDIA announced the dual-die B200 Blackwell GPU as a successor to their much sought after H100 and H200 Hopper GPUs. The new NVIDIA Blackwell B200 boasts a significant leap in computational power with over double the transistor count, 192 GB of HMB3e memory and 8 TB/s of bandwidth. The Blackwell B200 is a data-centre GPU and is expected to launch later in 2024. With startups such as Cerebras coming out with their WSE 3 'wafer scale' GPUs that have much higher computational power, cores, memory bandwidth and interconnect bandwidth than NVIDIA H100, it becomes imperative for NVIDIA to continue to push out bigger and better chips despite its market leadership.

NVIDIA's Blackwell GPU is named after Dr. David Harold Blackwell, an American statistician and mathematician who made significant contributions to probability theory, game theory and was also one of the three eponyms behind Rao-Blackwell-Kolmogorov theorem. NVIDIA hopes to continue the same momentum as the Hopper architecture and the Ampere architecture before it, with the introduction of the new Blackwell GPUs.

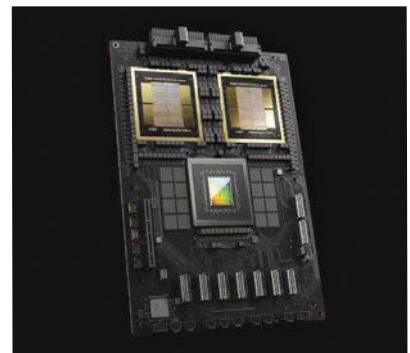
NVIDIA Blackwell goes the AMD way with MCM

With the Blackwell B200, NVIDIA has gone the AMD way of introducing chiplets in their design. The Blackwell B200 is actually built by slapping together two GPU tiles with 104 Billion transis-

tors, each. In comparison, the Hopper H100 had about 80 billion transistors. So the individual tiles have not scaled much but when you combine two tiles, you get a massive 208 billion transistors talking to each other using a 10 TB/s NV-HBI (NVIDIA High-bandwidth interface). NVIDIA Blackwell is built on TSMC's 4NP processor node which is a slightly improved version of TSMC's existing 4N node that's being used for the Hopper H100 GPUs. Also, the 4N node is a 5 nm process and not a 4 nm process.

The reason NVIDIA had to resort to using chiplets like AMD does with their Instinct MI300 GPUs is because the H100 GPUs were already nearing the reticle limits of TSMC's 4N process node. What this means is that the NVIDIA H100 GPU is one of the largest single GPUs that TSMC can make on a single silicon wafer. The H100 is 814 mm² in size whereas the reticle limit is about 858 mm². Therefore, scaling a chip requires the use of a multi-chip-module or MCM approach. And while NVIDIA Blackwell is a dual-chip GPU, NVIDIA does have something bigger planned to launch around 2025-26 called the GX200.

On the memory front, each Blackwell B200 die will use four HBM3e stacks of 16 GB that are connected to the GPU using a 1 TB/s interface. NVIDIA claims to have 20 petaflops of compute performance with the new Blackwell



NVIDIA Blackwell B200 Superchip